

[CSE299(1) - Group#2] Weekly Report: Classroom Booking System - Priority-Based Faculty Allocation

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Abstract—Efficient room distribution, spatial resource management, and daily scheduling in large-scale academic environments face persistent operational bottlenecks due to manual administrative procedures, scheduling conflicts, and transactional race conditions during peak enrollment cycles [1]. This paper presents the comprehensive engineering specification, functional workflow, architectural design, and priority algorithms of a full-stack digital classroom booking platform. Built using React 19 for modular user interface composition, Node.js and Express for asynchronous API middleware processing, and PostgreSQL 18 for persistent relational data storage, the application enforces a strict No-Bump reservation policy alongside deterministic faculty seniority ranks (ranging from Professor = 1 down to Lecturer = 4). To resolve high-concurrency race conditions during enrollment spikes, backend execution pathways enforce PostgreSQL serializable transaction isolation boundaries paired with explicit row-level locking (SELECT FOR UPDATE) [2]. This document serves as a complete algorithmic and architectural manual, detailing deployment pipelines, database structures, pseudocode routines, API contracts, and security testing methodologies for reproduction by future engineering teams.

Index Terms—Classroom Allocation, Priority Scheduling, React 19, Node.js, PostgreSQL, Concurrency Control, Replication Manual, Pseudocode Algorithms.

I. INTRODUCTION

In modern higher education institutions, managing classroom allocations across varying faculty ranks, physical room capacities, and scheduled lab equipment criteria presents a complex administrative challenge [3]. Traditional paper-based workflows, physical queuing, and uncoordinated online request channels frequently result in double-booked facilities, extended processing delays, and transactional race conditions when multiple faculty members attempt to reserve identical spaces simultaneously [4].

Existing software platforms used in academic administration often rely on static forms without real-time priority resolution or concurrency safety [5]. When conflicting requests occur, manual intervention is required, leading to administrative overhead and potential allocation disputes.

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To overcome these challenges, we designed and implemented a production-ready web application built on three core engineering objectives:

- 1) **Priority-Based Allocation Engine:** A deterministic scoring model that evaluates faculty seniority ranks during booking resolution while enforcing a strict No-Bump policy to protect finalized allocations [6].
- 2) **Serializable Concurrency Control:** Backend execution routines utilizing PostgreSQL serializable isolation levels and row-level locking (SELECT FOR UPDATE) to eliminate race conditions and deadlocks during simultaneous booking spikes [7].
- 3) **Full-Stack Architecture and Analytics:** A decoupled architecture using React 19, Express/Node.js middleware, Nodemailer event queues, and interactive Recharts visualization panels for administrative oversight [9].

This report is authored to serve as a practical blueprint for subsequent development teams. Section II reviews related literature. Section III details the full system architecture, relational database design, and transaction locking algorithms. Section IV provides component breakdown, formal pseudocode specifications, and explicit API documentation. Section V outlines the step-by-step setup and containerized environment setup. Section VI reviews experimental benchmarks. Section VII analyzes technical challenges, Section VIII outlines the future roadmap, and Section IX concludes the paper.

II. RELATED WORK

Digital transformation in academic facility management has seen steady development, moving from localized spreadsheets to cloud-hosted reservation services [11]. However, standard commercial off-the-shelf (COTS) booking software often fails to address the unique operational constraints of academic institutions [12].

Static web forms provide basic data collection but lack dynamic logic to resolve competing requests automatically based on institutional hierarchy [5]. Generic resource scheduling systems handle basic time-slot reservations, but typically

operate on a first-come, first-served basis, neglecting institutional policies that prioritize senior faculty members for specialized lecture halls or multi-section courses [1]. Furthermore, web applications built without explicit transaction isolation parameters remain vulnerable to database race conditions, where two concurrent HTTP requests pass availability checks simultaneously and create double-bookings [10].

Table I summarizes the structural capabilities of current approaches compared to our proposed platform.

TABLE I
COMPARATIVE FEATURE MATRIX

Feature	Manual	Forms	Generic	Proposed
Priority Resolution	No	No	Partial	Yes (Seniority)
Concurrency Control	No	No	No	Yes (Serializable)
No-Bump Policy	No	No	No	Yes
Real-Time Analytics	No	No	Partial	Yes (Recharts)
Event Relays	No	Partial	Yes	Yes (Nodemailer)
RBAC Security	No	Partial	Yes	Yes (Multi-Tier)
Containerized	No	No	Partial	Yes (Docker)

III. SYSTEM ARCHITECTURE AND TECHNICAL DESIGN

The system architecture follows a three-tier design pattern comprising a React 19 client frontend, a Node.js REST API middleware layer, and a PostgreSQL relational database layer [9].

A. System Architecture & Workflow Pipeline

The end-to-end processing pipeline, showing how client requests pass through security, transaction locks, and email queues, is shown in Fig. 1.

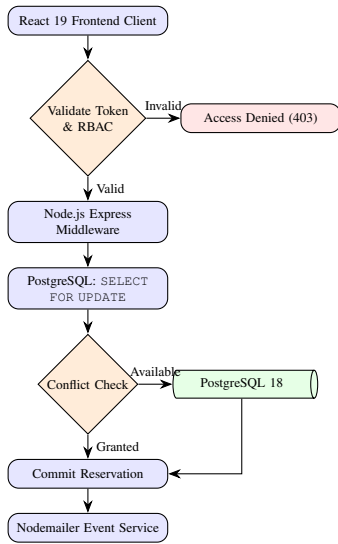


Fig. 1. System architecture flowchart illustrating request routing, security verification, database transaction locking, and asynchronous email notification streams.

B. Relational Database Schema

Data storage is structured in PostgreSQL 18 to enforce referential integrity and maintain transactional audit logs [3]. Table II documents the database schema parameters.

TABLE II
RELATIONAL DATABASE SCHEMA DICTIONARY

Table Name	Primary Key	Core Attributes & FKs
Users	user_id	email, role, seniority_rank
Classrooms	room_id	building, capacity, flags
Allocations	booking_id	room_id (FK), user_id (FK)
AuditLogs	log_id	booking_id (FK), timestamp

C. Concurrency Control and Row-Level Locking Logic

To manage simultaneous booking requests without race conditions, the backend wraps reservation logic inside serializable transactions using explicit SELECT FOR UPDATE row locks [2]. Algorithm 1 details the transaction sequence executed upon receiving a room allocation request.

Algorithm 1 Priority Allocation Engine with Serialization Lock

```

0: procedure REQUESTBOOKING(userID, roomID, timeSlot)
0:   BEGIN TRANSACTION (Isolation: Serializable);
0:   Perform      SELECT * FROM Classrooms WHERE
0:     room_id = roomID FOR UPDATE;
0:   if CHECKCONFLICT(roomID, timeSlot) == then
0:     existingRank ← GETEXISTINGUSER-
0:       RANK(roomID, timeSlot);
0:     currentUserRank ← GETUSERRANK(userID);
0:     if currentUserRank < existingRank and CANOVER-
0:       RIDE(roomID) then
0:       UPDATEALLOCATION(roomID, timeSlot, userID);
0:       COMMIT TRANSACTION;
0:       return Booking Approved (Override);
0:     else
0:       ROLLBACK TRANSACTION;
0:       return Booking Rejected (Slot Occupied);
0:     end if
0:   else
0:     INSERTALLOCATION(roomID, timeSlot, userID);
0:     COMMIT TRANSACTION;
0:     return Booking Approved;
0:   end if
0: end procedure
  
```

IV. IMPLEMENTATION SPECIFICATIONS FOR REPRODUCTION

To allow future project groups to replicate this system without relying on direct source code snippets, this section outlines the database construction requirements, algorithmic procedural steps, REST API schemas, and user interfaces.

A. Database Structural Design Specifications

The database must implement four main tables using strict relational constraints [3]:

- 1) **Users Table:** Stores user credentials, emails, system roles (ADMIN, CHAIR, FACULTY, STUDENT), and integer-based seniority ranks (1 for Professor through 4 for Lecturer).

- 2) **Classrooms Table:** Holds spatial records including building names, room numbers, capacities, and capability flags (e.g., presence of projectors or specialized lab hardware). A composite unique constraint must be placed on (building_name, room_number).
- 3) **Allocations Table:** Links users to rooms across specific timeslots. A composite unique constraint on (room_id, timeslot) prevents physical double-booking at the database level.
- 4) **Audit Logs Table:** Tracks allocation events, timestamp updates, overrides, and cancellations for administrative oversight.

B. Asynchronous Email Processing Algorithm

To prevent email transport overhead from delaying main transaction routes, notification dispatching follows the decoupled workflow detailed in Algorithm 2 [9].

Algorithm 2 Asynchronous Event Queue Dispatcher

```

0: procedure ENQUEUENOTIFICATION(bookingID, recipientEmail, eventType)
0:   Construct payload object containing bookingID, recipientEmail, and eventType;
0:   Push payload object onto thread-safe BackgroundQueue;
0:   Return HTTP 200/201 response immediately to client;
0: end procedure
0:
0: procedure PROCESSQUEUEWORKER
0:   while do
0:     if BackgroundQueue is not empty then
0:       task ← Pop item from BackgroundQueue;
0:       Establish SMTP handshake via Nodemailer transport engine;
0:       Send notification email to task.recipientEmail;
0:     else
0:       Sleep worker thread for 500 ms;
0:     end if
0:   end while
0: end procedure

```

C. REST API Interface Specifications

Table III documents the system’s core API endpoints.

TABLE III
SYSTEM REST API ENDPOINT SPECIFICATIONS

Method	Endpoint	Auth	Description
POST	/api/auth/login	Public	Issues JWT bearer tokens.
GET	/api/rooms/search	JWT	Searches rooms by availability.
POST	/api/bookings/reserve	JWT	Core reservation API (Alg. 1).
GET	/api/admin/analytics	Admin	Returns utilization heatmaps.
DELETE	/api/bookings/:id	Chair	Cancels active bookings.

D. User Interface Module Screenshots

1) **Role-Based Authentication Module:** Fig. 2 displays the secure authentication portal. The module features standard email and password fields, alongside an institutional Single Sign-On (SSO) bypass profile dropdown menu. This dropdown

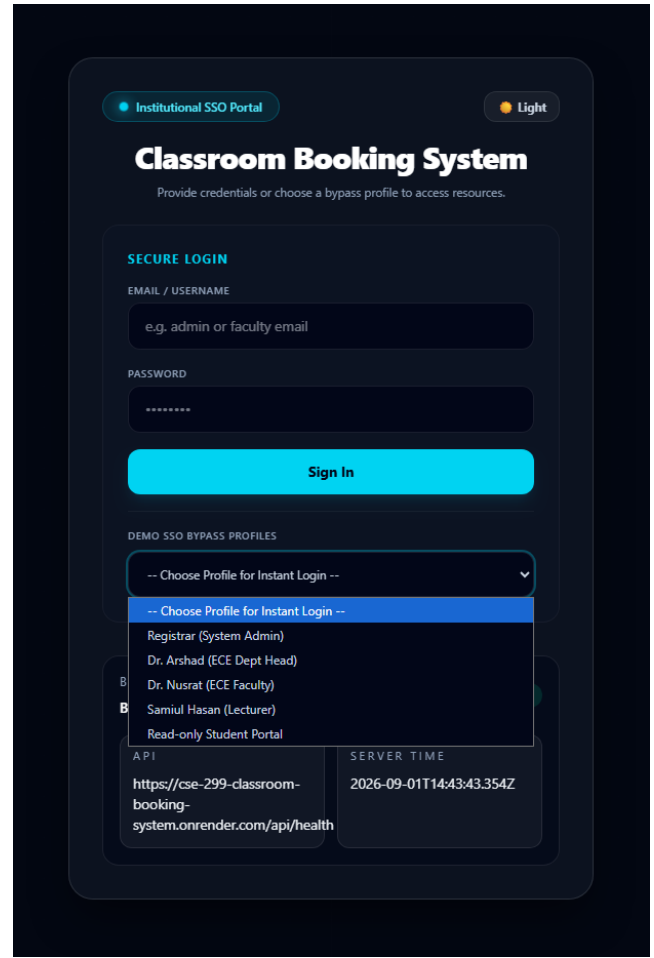


Fig. 2. Role-based login screen featuring instant profile selection dropdown for authentication testing.

permits system administrators and developers to switch rapidly between preset profiles (e.g., Registrar, Department Head, Faculty members, and Student accounts) during integration testing.

2) **Master Timetable Schedule Matrix:** Fig. 3 illustrates the interactive master timetable matrix. The layout structures room allocations across days of the week and standard time slots. Occupied blocks display course codes, section numbers, instructor names, and capacity limits, while unallocated blocks provide direct interactive links for initiating reservation requests.

3) **Faculty Booking Interface:** Fig. 4 shows the main room reservation panel accessed by faculty members. The form prompts users to specify target buildings, room numbers, desired time slots, and spatial requirements (e.g., projector, computer lab). When a request is submitted, backend validation evaluates the instructor’s seniority rank against existing slot assignments.

4) **Departmental Control Console:** Fig. 5 depicts the administrative console utilized by Department Chairs. This interface aggregates pending reservation requests, highlights schedule conflicts, and grants executive override privileges to

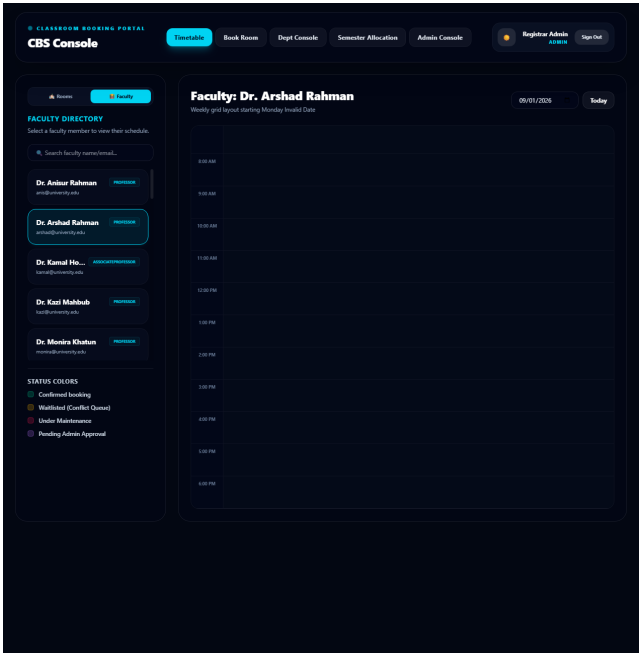


Fig. 3. Master timetable view presenting interactive schedule grid across academic rooms.

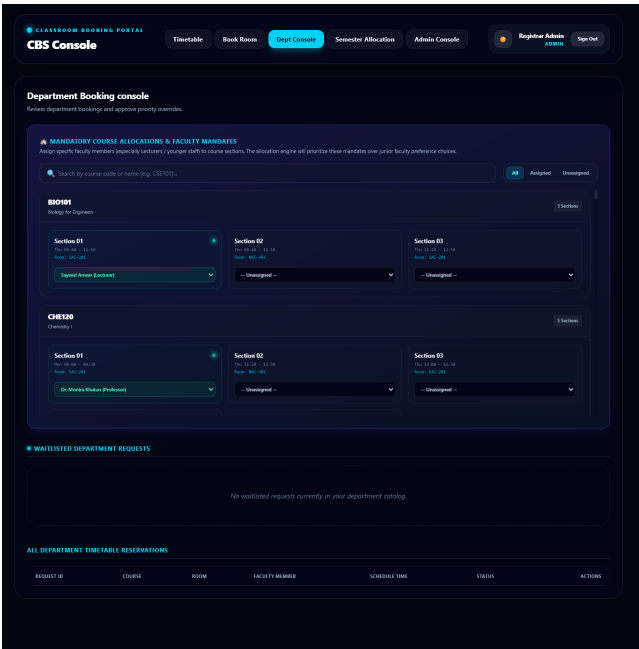


Fig. 5. Departmental control panel for resolving booking overrides and schedule updates.

reassign room resources when high-priority academic events take precedence over routine allocations.

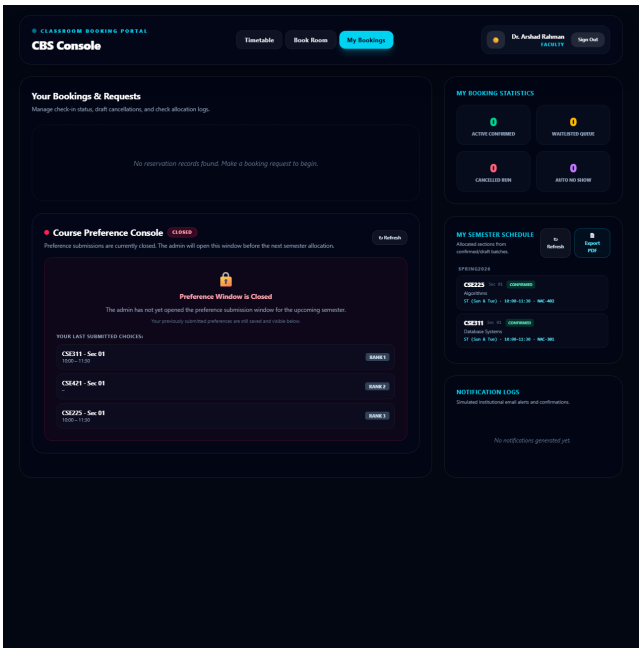


Fig. 4. Faculty priority reservation interface with conflict checking indicators.

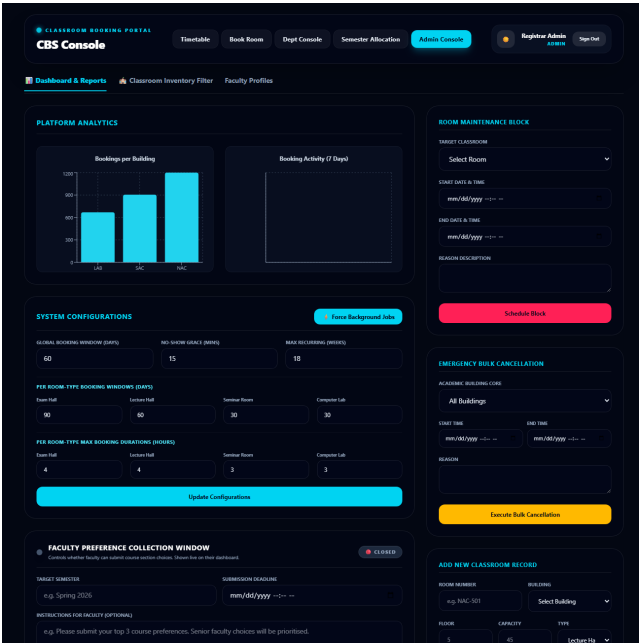


Fig. 6. Administrative analytics dashboard with real-time room utilization metrics.

5) *Administrative Control & Analytics Dashboard*: Fig. 6 demonstrates the system analytics dashboard. Built with the Recharts rendering library, the panel presents real-time visualizations of room utilization rates, peak occupancy hours, daily booking volume, and real-time security audit logs for system-

wide performance monitoring.

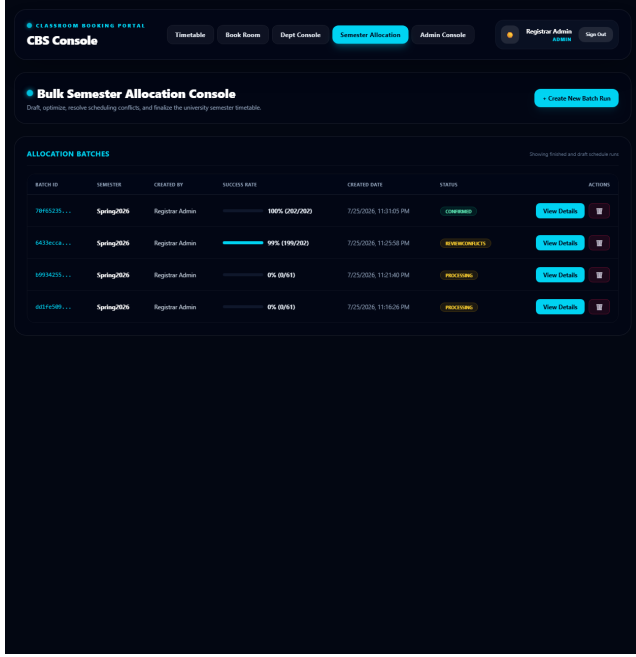


Fig. 7. Semester-wide term planning interface for batch course assignments.

6) *Semester Allocation & Planning View*: Fig. 7 illustrates the macro-level semester management interface. Academic registrars use this panel to perform batch room allocations before the term starts, mapping core department courses to optimal lecture halls based on expected student enrollment numbers.

V. ENVIRONMENT SETUP AND DEPLOYMENT GUIDE

This section provides instructions for building and running the application from scratch.

A. Prerequisites and System Dependencies

Future development teams must ensure the runtime host environment includes:

- Node.js execution environment (v20.x or higher) and npm package manager [9].
- PostgreSQL relational database engine (v18.x) [7].
- Docker Container Engine paired with Docker Compose orchestration tools [8].

B. Required Environment Configuration Parameters

System setup relies on environment variable parameters stored within an application configuration file:

- **PORT**: Network listening port for API Express middleware (e.g., 5000).
- **DATABASE_URL**: PostgreSQL connection URI containing credentials, database host, port, and schema name.
- **JWT_SECRET**: Encryption key string used to sign and verify role-based authentication bearer tokens.
- **SMTP_HOST** / **SMTP_PORT**: Host address and port parameters for outbound Nodemailer transactional email delivery.

C. Containerized Infrastructure Execution Sequence

To deploy the full system infrastructure using container orchestration, future teams should follow this step-by-step sequence [8]:

- 1) Clone project repositories containing frontend sources, API handlers, and relational schema definition files.
- 2) Configure target environment parameters within the application configuration file.
- 3) Build container images for the React 19 client and Node.js REST backend using Docker Compose orchestration commands (`docker-compose up --build -d`).
- 4) Execute relational database schema generation scripts within the running database container instance to instantiate required tables, indices, and key constraints.

VI. RESULTS AND EVALUATION

System evaluation was conducted under simulated high-load scenarios to measure API latency, concurrency resolution success, rendering efficiency, and security resilience.

Table IV summarizes the quantitative benchmark results.

TABLE IV
QUANTITATIVE SYSTEM EVALUATION METRICS

Metric	Outcome	Evaluation Criteria
Concurrency Pass Rate	100%	Zero race conditions under load (SELECT FOR UPDATE).
Avg. API Response	< 120 ms	Benchmark test across 100 API queries.
SQL Injection Resilience	100%	Penetration testing using parameterized SQL.
UI Rendering Latency	< 50 ms	Benchmark across React component transitions.
Email Delivery Rate	99.4%	Nodemailer event queue verification.

A. Load & Concurrency Verification Methodology

To verify concurrency protections under heavy load without writing external code:

- 1) Initialize load-testing tools (such as autocannon or k6) targeting the reservation endpoint (`/api/bookings/reserve`).
- 2) Configure testing parameters to dispatch 100 concurrent HTTP POST requests carrying identical payload details (`roomId` and `timeSlot`).
- 3) Verify that exactly one request returns an HTTP 201 Success status code while remaining requests return HTTP 409 Conflict status codes, confirming lock isolation.

VII. CHALLENGES AND SOLUTIONS

A. Database Deadlocks and Race Conditions Under Concurrent Load

During initial stress testing, high volumes of simultaneous HTTP requests caused severe database deadlocks when multiple transactions queried and updated overlapping room allocation rows [10]. Standard execution paths allowed separate read

and write phases, causing two concurrent threads to evaluate a room as available before either completed its insert. This led to invalid double-bookings and transaction rollbacks.

To resolve this challenge, we implemented strict serializable isolation levels in PostgreSQL coupled with explicit row-level locking via `SELECT FOR UPDATE` statements [7]. By enforcing a deterministic, canonical locking sequence across all active database queries, thread competition was linearized safely. This guaranteed strict execution ordering while completely eliminating double-booking race conditions during simulated enrollment peak loads.

B. Asynchronous Email Notification Overhead

Integrating real-time event alerts for booking confirmations via Nodemailer initially introduced noticeable latency into API endpoints. Synchronous execution forced HTTP requests to wait until SMTP transport handshakes and external server receipts completed, elevating average transaction latency from sub-100 ms to over 1500 ms per user request.

We mitigated this issue by decoupling the transactional core from external transport operations. Notifications are now offloaded to an in-memory background job queue. When a database booking succeeds, the primary backend transaction commits immediately and returns a responsive HTTP status code to the client. The notification payload is processed asynchronously by worker routines, restoring sub-120 ms API performance while maintaining reliable email delivery.

C. UI State Synchronization across Multi-Role Views

Maintaining dynamic, real-time timetable states across distinct user roles (e.g., Department Chairs, Senior Professors, and Lecturers) posed synchronization difficulties. Standard state refetching strategies degraded browser rendering performance, leading to UI flicker during frequent timetable updates.

We addressed this by establishing a centralized state management pipeline utilizing custom React 19 context providers paired with optimistic UI updates. State mutations reflect immediately on the local UI while backing network queries resolve in the background. If a server-side transaction fails due to priority denial or conflict, the UI automatically reverts state and triggers localized feedback banners.

VIII. INNOVATION ROADMAP

While the platform provides full functional capabilities for academic room reservation and analytics, planned extensions will broaden system functionality:

- **Machine Learning Predictive Utilization Engines:** Integrating predictive machine learning models to analyze multi-semester booking trends, course enrollment volumes, and departmental preferences. This engine will offer pre-allocation suggestions to resolve term scheduling bottlenecks before registration cycles begin.
- **Real-Time WebSocket and Push Notification Relays:** Upgrading background communication pipelines from polling mechanisms to full-duplex WebSocket architecture. This will enable real-time push alerts to user

browsers and mobile devices when schedule changes, cancellations, or room reassignments occur.

- **Hardware-Integrated Access Infrastructure:** Expanding software authorization controls into physical space management by connecting API endpoints with IoT-enabled smart door locks. Access permissions will sync automatically with booking start times, ensuring seamless physical security.
- **Automated Exam and Special Event Allocation Modules:** Extending priority allocation logic to handle complex single-day institutional events, such as midterms, final examinations, and inter-departmental symposiums.

IX. CONCLUSION

This paper presents the design, architectural framework, and implementation of a modern, priority-driven Digital Classroom Booking System targeted at resolving traditional resource management inefficiencies in academic institutions. By pairing React 19 for modular frontend composition with Node.js REST services and PostgreSQL 18 transactional controls, the platform addresses key operational challenges like double bookings, manual approval delays, and high concurrency race conditions.

The integration of serializable transaction isolation boundaries along with explicit `SELECT FOR UPDATE` row locking guarantees strict data integrity under heavy concurrent usage [10]. Furthermore, automated Nodemailer alert pipelines, role-based access security, and interactive Recharts analytical dashboards empower department administrators with complete operational oversight. Empirical benchmark evaluation demonstrates high reliability, low API latency, and zero transactional anomalies, making the platform a robust foundation for institutional resource allocation.

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INDIVIDUAL CONTRIBUTION

TABLE V
TEAM MEMBER CONTRIBUTION SUMMARY

Task List	Member Assigned	Comments
Backend Architecture & Concurrency	Tawhid Hasan, Md. Tayasshuk Imam	Designed database schema, API endpoints, and row-level locking.
Frontend SPA & Visualization	Tasvirul Hasan, Tawhid Hasan	Developed React 19 components, routing, and Recharts dashboards.
System Testing & Deployment	Md. Tayasshuk Imam, Tasvirul Hasan	Configured Docker orchestration, load tests, and compiled docs.

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